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Maghreb's renewable power future for climate mitigation: insights from the TIMES-MAGe model

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ABSTRACT

The Maghreb's power-sector transition hinges on implementing the region's Nationally Determined Contributions and Low-Emission Development Strategies. Using The Integrated Market Allocation Energy Flow Optimization Model system (TIMES) energy-modelling framework, this study traces the evolution of the electricity systems of Tunisia, Morocco, and Algeria from 2018 to 2050, assessing renewable energy targets, decarbonization feasibility, and the value of regional electricity trade. All three countries can exceed their renewable electricity targets: Onshore wind and solar photovoltaic jointly comprise over 90 % of the capacity, while Concentrated solar power plays a limited role. Flexible technology deployment is expected to result in a 25 % reduction in electricity prices in Morocco and Tunisia, and a 10 % reduction in Algeria, by 2030. Power sector decarbonization requires 130 GW of renewable energy capacity and 28 GW of storage, with a cumulative investment of €17 billion. Regional electricity trade enables cost savings, reducing Algeria's investment needs by 18 %, and allows Morocco and Tunisia to export up to 71 % and 54 % of their interconnector capacity. These results underscore the importance of regional cooperation and provide valuable insights for aligning national energy strategies with long-term climate objectives.

List of abbreviations, including units and nomenclature

AA	Accelerated ambitious		
BRICS	Brazil, Russia, India, China, and South Africa	**NaS**	Sodium–sulfur batteries
CESA	Continental European synchronous area	**NDCs**	national determinant contribution
CF	Capacity factor	**O&M**	Operations and Maintenance
CSP	Concentrated solar power	**PA**	Paris Agreement
DZR	Algeria	**PAERTE**	Project for the development and equipment of the electricity transmission network
EDGAR	Emissions Database for Global Atmospheric Research	**PDRTE**	Power transmission and distribution network development project
ESM	Energy system model	**PJ**	Petajoule
ESIP	Energy sector improvement project	**PP**	Power plant

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ETSAP	Energy Technology Systems analysis program	**PST**	Tunisian solar plan
GDP	Gross domestic product	**PV**	Photovoltaic
GHG	Greenhouse gas emissions	**RES**	Renewable energy sources
GIS	Geographic information system	**RES-E**	Renewable Energy Sources – Electricity
GW	Gigawatt	**RISE**	Regulatory indicator for sustainable energy
IPCC	The Intergovernmental Panel on Climate Change	**STEG**	Tunisian company of electricity and gas
IRENA	International Renewable Energy Agency	**TEASIMED**	Towards an Efficient, Adequate, Sustainable, and Interconnected Mediterranean
Li-On	Lithium-ion batteries	**TFC**	Total final consumption
LT-LEDS	Long-term low-emission development strategy	**TIMES**	The Integrated MARKet ALlocation – Energy Flow Optimization Model system.

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LEAP	The Low Emissions Analysis Platform	**TUM**	Technical University of Munich
LUT	Lappeenranta University of Technology	**TUN**	Tunisia
OPEC	The Organization of the Petroleum Exporting Countries	**TU Wien**	Vienna University of Technology
ONEE	National Office of electricity and water	**UNFCCC**	United nations framework convention on climate change
MAR	Morocco		
MEM	Maghreb electricity market		
MENA	Middle East and North Africa		

1. Introduction

The global imperative to mitigate climate change has intensified the need to transform energy systems, particularly the power sector, which is central to the decarbonization of all other sectors [1,2]. According to the Intergovernmental Panel on Climate Change (IPCC), reaching net-zero emissions by mid-century requires rapid deployment of low-carbon technologies and a significant shift in electricity generation toward renewable sources. By 2050, electricity systems worldwide must integrate 70–85 % renewables [3], which demands more than isolated technological improvements: it requires coordinated national strategies supported by coherent short-, medium-, and long-term policy frameworks. The Paris Agreement[1] and its mechanisms, particularly Nationally Determined Contributions (NDCs) and Long-Term Low Emission Development Strategies (LT-LEDS), offer countries both structure and flexibility to design these transitions [4]. However, aligning these instruments in a manner that is both cost-efficient and technically feasible, while also being socially acceptable, remains a significant challenge.

While NDCs and LT-LEDS have different features, they are complementary. NDCs represent the country's self-determined near-term targets and actions for reducing its GHG emissions over a specific timeframe, usually five years. Conversely, LT-LEDS establishes a long-term strategic framework for countries' transition towards a low-carbon economy. LT-LEDS adopts a comprehensive, forward-looking approach that extends beyond the immediate targets of NDCs, identifying a range of actions and investments that can effectively reduce GHG emissions while promoting sustainable development.

In this context, The Maghreb[2] region, comprising Morocco, Algeria, and Tunisia, stands at a critical juncture. Although endowed with abundant solar and wind resources [5], these countries remain primarily dependent on fossil fuels for electricity generation [6]. Renewable energy penetration remains limited, and electricity trade among the three countries is minimal despite political discourse around regional cooperation. Meanwhile, domestic electricity demand is rising, infrastructure is aging, and national climate ambitions are evolving. Each country has updated its NDC and proposed ambitious targets for renewable electricity, yet how these ambitions translate into long-term system-level change remains poorly understood. Morocco is a leader in renewable development, targeting 80 % [7] Renewables by 2050. Tunisia aims for 35 % by 2030 [8], with efforts to accelerate deployment. In contrast, Algeria lags due to its reliance on natural gas and limited integration of climate policy [9].

The adoption of LT-LEDS across Maghreb countries marks a critical step in extending climate planning beyond near-term NDCs [10]. Yet, these strategies remain largely absent from energy system analyses, leaving their implications for national power transitions unexplored [11]. In addition, their country-by-country framing overlooks cross-border dynamics, a significant gap in the Maghreb, where regional electricity trade could impact investment pathways and system reliability. This study addresses these omissions by examining how national LT-LEDS targets interact with opportunities for intra-Maghreb cooperation. In contrast, energy and climate research on North Africa and the wider MENA region is expanding [12]. Most studies adopt broad regional perspectives (e.g, Refs. [13–15]) that obscure country-specific priorities and overlook links between short-term pledges, long-term strategies, and prospects for full decarbonization. Moreover, although EU–MENA integration has been widely studied, the dynamics within the Maghreb region remain largely neglected. These gaps limit understanding of how Maghreb countries can reconcile national ambitions with collective pathways toward a reliable and cost-effective power transition.

This paper addresses these shortcomings by developing and applying the TIMES-MAGe model to examine the potential for renewable electricity transitions in the Maghreb. TIMES-MAGe model is a novel techno-economic optimization framework that models the three countries to assess renewable energy targets, decarbonization feasibility, and the value of regional electricity trade by 2050. By linking updated NDCs and long-term targets to investment and technology deployment trajectories, the study offers new insights into the role of renewables and regional integration in supporting climate mitigation in Africa Maghreb countries. It aims to answer.

- To what extent do the planned capacities in Maghreb countries fulfil the renewable electricity targets?
- Is there potential for optimizing predefined capacity plans to enhance the cost-effectiveness of achieving national RES goals in the Maghreb countries?
- To what extent can the Maghreb region reach full power sector decarbonization (100 % RES-E), considering technological capabilities and resource potential?
- What are the necessary costs, including investments, operations, maintenance, and net import needs, and the resulting electricity unit costs, under these scenarios?
- What are the trade-offs of enabling regional electricity trade in the Maghreb?

The paper is organized as follows: Section 2 reviews the relevant literature on energy system modeling and outlines the study's contribution. Section 3 examines the Maghreb's electricity systems and renewable targets within NDCs and LT-LEDS. Section 4 outlines the methodology, which includes the TIMES framework, model development, and scenario analysis. Section 5 presents comparative results on capacity, generation, investments, costs, and trade. Section 6 concludes with key findings and policy insights.

2. Literature review

Energy system models (ESMs) have long supported climate policy design by providing decarbonization pathways and quantifying their implications for decision-makers [16]. They have been widely applied across diverse contexts, including Europe [17–20], North America [21–23], Asia [24–27], and Sub-Saharan Africa [28–31]where they inform long-term strategies, guide investment planning, and assess the feasibility of achieving net-zero targets.

[1] The Paris Agreement is an international treaty on climate change that was adopted in 2015. The treaty covers climate change mitigation, adaptation, and finance. The Paris Agreement was negotiated by 196 parties at the 2015 United Nations Climate Change Conference near Paris, France.

[2] The Arab Maghreb Union is a political union and economic union trade agreement aiming for economic and future political unity Its members are the nations of Algeria, Libya, Mauritania, Morocco, and Tunisia however, this research is limited only to the three major countries namely Tunisia Morocco and Algeria. Libya and Mauritania were left out given the different conflict setting and political instability.

Despite these advances, the application of ESMs to the Maghreb remains limited. Most studies embed the Maghreb within broader MENA-focused analyses, diluting attention to its distinct challenges. Huber et al. [13] utilized the TUM-ESM to explore cost-effective electricity configurations in the MENA region, revealing substantial cost savings from cross-regional electricity exchange, primarily driven by wind power. Similarly, Aghahosseini et al. [14] applied the LUT -ESM to examine the feasibility of a fully renewable MENA system by 2030, emphasizing the dominance of solar PV and wind. Fragkos [15], also, using MENA-EDS, underscored the value of EU-MENA energy cooperation in driving decarbonization, while a non-peer-reviewed study [32] combining Powerace, GIS, and Green_X models projected a near-total RES-E network by 2050 and highlighted the importance of integrating North Africa through PV, CSP, and onshore wind.

Together, these studies emphasize that EU engagement can catalyze low-carbon transformation in the Maghred, aligning emissions reductions with geopolitical and economic interests. Yet, their regional lens often overlooks critical intra-regional differences. Similarly, Brand [33]Using the DIME model, the study explored how conservative 2025 RES-E targets in Morocco, Algeria, and Tunisia could partially replace fossil fuels. However, the short-term focus and modest ambitions fail to reflect the evolving national climate goals.

Moreover, as renewable penetration increases, the complexity of energy transitions in the Maghreb hinges not only on physical interconnections but also on secure digital infrastructures. For instance, recent Internet of Energy (IoE) frameworks propose semi-decentralized architectures where energy routers manage dispatch and enhance cyber-physical security [34]. While not power-sector specific, such advances highlight the growing role of digital layers in ensuring reliable integration of variable renewables. Beyond electricity generation itself, synergies with other sectors further shape long-term transition pathways. Vehicle fleet decarbonization studies [35], for example, indicate that large-scale adoption of electric and hydrogen vehicles in the Maghreb could substantially increase electricity demand while providing flexibility and storage opportunities. Although these dynamics extend beyond the strict boundaries of the power sector, they underscore the importance of considering systemic interactions when modelling electricity transitions.

Prior work on the Maghreb's power transition has been divided into two strands. Regional system-integration studies examine high-RES futures and grid operation at multi-country scale. For example, Boie et al. [36] Combine PowerAce, RESlion, and DIgSILENT to demonstrate that near-100 % renewable portfolios are technically viable across five North African systems, with grid reinforcements and efficiency measures reducing electricity costs by more than one-third. Yet the regional lens tends to blur national constraints, policy regimes, and investment priorities. Complementing this, national case studies illuminate country-specific pathways: in Tunisia, OSeMOSYS analyses indicate decentralized PV as a long-term cornerstone and the need for stronger policies to align with theParis Agreement [37,38], in Morocco, optimization and scenario studies find that phasing out coal and scaling RES, supported by flexibility options, stabilizes costs and advances NDC/LT-LEDS consistency, while policy syntheses summarise the enabling framework [39,40], techno-economic work further tests feasibility under rising demand and identifies conditions for 100 % RES cases and cost-effective hybrid PV-wind solutions, alongside demand-side efficiency roadmaps [41,42]. By contrast, Algeria remains comparatively under-modelled: TIMES-DZA and OSeMOSYS studies point to trajectories centred on solar (and CSP with storage) and onshore wind toward mid-century, but often omit explicit cost and investment dynamics, limiting policy relevance [43,44]. A more recent LEAP-based study extends this work by incorporating investment costs and exploring the interplay between hydrogen exports and domestic decarbonization [45]. Yet, this analysis remains country-specific and centred on hydrogen policy, whereas broader system-level assessments that situate Algeria within the regional dynamics of the Maghreb power transition are still lacking, a gap this study seeks to address.

Across these strands, three gaps persist. First, most analyses prioritize 2030 NDCs and short-term targets [38,39,41,46–48], with limited integration of LT-LEDS and scant assessment of how near-term pledges align with long-term decarbonization; to date, only Morocco has a dedicated LT-LEDS review [49]. Second, country specificity is frequently diluted in regional models, while Algeria is underrepresented. Third, many studies adopt coarse temporal resolution, under capturing RES availability and demand variability that shape system integration and costs [50–52].

This paper responds to the gaps above by developing and applying a regionally interconnected, country-specific TIMES model that simulates renewable electricity transitions in Morocco, Tunisia, and Algeria through 2050. Integrating national NDCs and LT-LEDS, our model tests deployment and cooperation scenarios, as well as 100 % renewable scenarios, to assess whether current renewable plans align with long-term targets and how different strategies affect costs, investments, and reliability. By comparing national self-reliance with regional cooperation, we provide a comprehensive economic and technical assessment of decarbonizing the Maghreb's power sector.

Our contribution lies in developing a replicable, regionally integrated energy modeling approach, TIMES-MAGe, for academics and as a decision-support tool for policymakers to identify optimal trajectories and cooperation needs. This study presents a robust methodological approach informed by an extensive policy review and an examination of the current technology landscape in the Maghrebian power sector.

3. Context

The Maghreb, comprising Tunisia, Algeria, and Morocco, counted 96.3 million inhabitants in 2023, a figure projected to rise by 14 % by 2030 and 26 % by 2050 [53]. Despite its abundant renewable potential, the region's energy system remains dominated by fossil fuels, with natural gas supplying nearly two-thirds of primary consumption [54]. Algeria's economy is particularly exposed, with gas exports representing 19 % of GDP and 93 % of total exports in 2022 [55]. Electricity accounts for one-fifth of final energy demand yet generates a quarter of regional CO_2 emissions [54]. Demand growth, coupled with aging infrastructure, has strained power systems: in 2018, enterprises faced an average of 120 lost working hours per month due to outages, eroding sales by 10 %, twice the global average [56,57].

Energy dependence profiles further underscore structural asymmetries (Table 1). Morocco imports around 90 % of its energy supply, Tunisia 55 %, while Algeria is a net exporter (−8.4 %) [58]. This fossil-fuel dependence exposes Morocco and Tunisia to price volatility and supply shocks, while Algeria's export reliance makes it highly sensitive to shifts in global demand. Together, these vulnerabilities underscore the urgency of accelerating a sustainable power sector transition in the Maghreb.

Table 1
Primary energy consumption, share of electricity in final energy consumption, installed capacities, and energy dependence rate in the Maghreb countries in 2023 [59–63].

		Tunisia	Morocco	Algeria
Primary energy consumption (PJ)	Natural gas	185.6	8.85	2165
	Coal	0	287	2.5
	Oil	70.73	497	933
	Renewables	38.7	54.5	10
Share of electricity in final energy consumption (%)		18	19	13
Total power installed capacities (GW)		6.3	11	22.6
Energy independence rate (%)		55	90	−8.4
Electricity consumption/capita (MWh/year/cap)		1.55	0.95	1.75

3.1. The Maghreb's power sector

The Maghreb has undergone significant economic and industrial growth in recent decades [63], leading to a substantial increase in electricity demand (Fig. 1) an annual average growth rate of around 2.87 % between 2010 and 2020 [64].

The Maghreb's power sector remains overwhelmingly fossil-dependent (Fig. 2), with natural gas and coal supplying 93 % of electricity in 2023 [65–67]. Tunisia and Algeria rely almost entirely on gas (96 %), while Morocco depends mainly on coal (70 %). Renewables contributed just 17 % regionwide, led by Morocco (42 % of its mix), followed by Tunisia (11 %) and Algeria (<1 %). In terms of capacity, renewables represent only 8 % of the regional total [68]. Morocco dominates with 5 GW, including the Noor Ouarzazate CSP, the world's largest [69], while Tunisia and Algeria together add just 1.4 GW. Yet solar and wind potential across the region remains substantial [68,70]. Policy commitments point to rapid growth. Morocco targets a 10 GW expansion by 2030 [7], Tunisia has awarded 1.7 GW of PV for delivery in 2025–2026 [71], and Algeria plans 15 GW of solar by 2035, starting with an 80 MW plant in Bechar [71]. These developments signal a gradual but critical shift toward renewables in the Maghreb's electricity mix.

Electricity trade in the Maghreb remains limited, at just 13 PJ in 2020, less than 4 % of regional consumption [73]. Algeria supplies most cross-border flows, notably 12 PJ to Tunisia in 2023, while exchanges with Morocco have declined amid diplomatic tensions [73,74]. Tunisia also exports to Algeria and Malta, though technical issues constrain the Malta undersea cable [75]. Regional integration efforts, including the Maghreb Electricity Market (MEM) launched in 2013 and a 2014 bilateral agreement enabling limited trade, have yielded little progress. In 2023, national utilities (SONELGAZ, ONEE, STEG) began negotiating to ease cross-border constraints and explore direct Tunisia–Morocco exchanges via Algeria, though these remain unrealized [76,77]. Currently, Morocco–Spain interconnection dominates, accounting for 76 % of Morocco's cross-border electricity trade [77]. Looking ahead, the establishment of new Mediterranean links, such as Algeria-Italy (2027) and Tunisia–Italy (2025), could significantly expand regional and Euro-Maghreb power integration.

3.2. Climate mitigation and renewable targets

The Maghreb countries show divergent climate ambitions and RES targets (Table 2). Morocco targets a 45.5 % GHG reduction by 2030 (BAU), with 52 % RES capacity in 2030 and 80 % by 2050, including 4.9 GW wind, 5 GW solar, and 1.4 GW hydropower [7,8,68,70]. Tunisia aims for a 45 % cut in emission intensity, 30 % RES by 2030, and 80 % by 2050, with 4.6 GW RES by 2030 (2.6 GW PV, 1.6 GW wind, 0.2 GW CSP, 0.1 GW biomass). Algeria's previous 2030 targets ranged 7–22 % reduction, with current plans for 27 % RES and 22 GW capacity, dominated by solar (13.5 GW) and wind (5 GW) [68]. These milestones illustrate the evolving trajectories and divergence of national energy strategies in Morocco, Tunisia, and Algeria, providing context for the current and planned renewable energy capacities (Fig. 3).

RES technologies expansion targets and required capacities expansion for 2030 and 2050, for the Maghreb countries, are regrouped in Table 2.

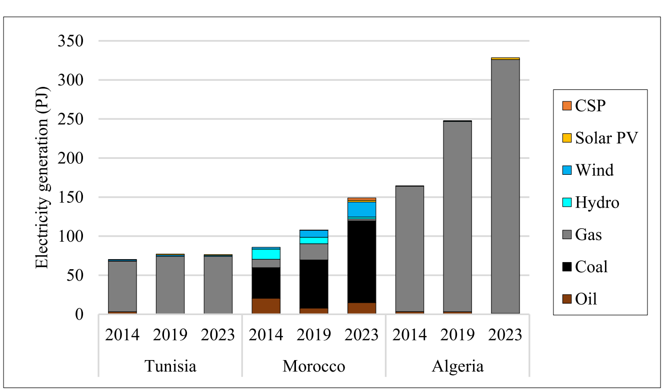

Fig. 1. Electricity generation by source in the Maghreb Countries.

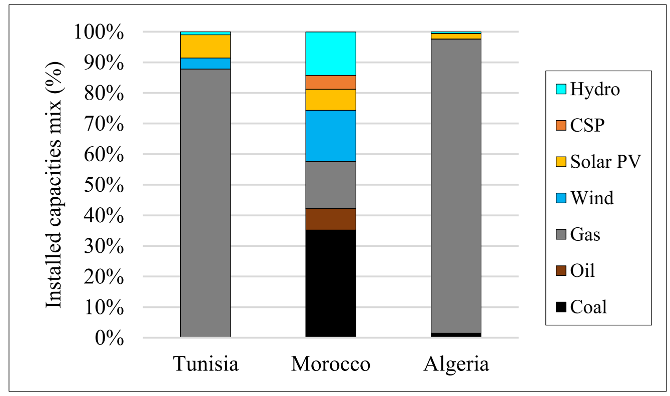

Fig. 2. Share of power installed capacity per technology in the Maghreb countries in 2023 [72].

4. Methodology

4.1. Overview of the TIMES model

TIMES (The Integrated MARKAL-EFOM System) model is a widely used [71] Energy system modelling tool, employed to explore energy transition scenarios and identify cost-effective mitigation pathways, which can be applied to several energy systems scales (i.e., global, regional, and local) [81,82]. Detailed mathematical modelling of the TIMES framework can be found in Refs. [83,84]. The model has been widely applied in different countries, informing decarbonization pathways (e.g., Egypt [85]), emphasizing cross-border integration (e.g., Portugal [86]), exploring emission reduction scenarios (e.g., Turkey [87]), and assessing renewable energy prospects (e.g., Malaysia [88]), thus demonstrating its broad applicability.

4.2. TIMES-MAGe

The cost-effectiveness of investments in renewable energy in the Maghreb is being assessed through the development of the first Maghreb TIMES model (Fig. A1). The TIMES-MAGe covers the power sectors of the three Maghreb countries: Morocco, Tunisia, and Algeria, running from 2018 (the base year) up to 2050 in two and then five time periods, i.e., 2018, 2020, 2025, 2030, …up to 2050. Fig. 4, illustrates the overall scope and structure of this study. It highlights the TIMES-MAGE model boundaries, country coverage, scenarios analysed, and key outputs.

The model is calibrated in euros (€) with 2018 as its base year, chosen for its comprehensive and harmonized datasets across Tunisia, Morocco, and Algeria. Key capacity inputs were updated to 2023 values and validated against observed 2023 electricity demand, achieving deviations of no more than 3 %. All cost data remain expressed in 2018 EUR; to convert to 2023 EUR, an inflation factor of 1.22 is applied [89].

To capture the variability of renewable capacity factors and electricity demand, the model considers 48 time slices. Further subdivided into representative days (1 day per each of the 4 seasons) and hours (each day is modelled in 2-h segments: 0–2h, 2h–4h, 4h–6h … 22-0h). Using these time slices ensures that the model can match electricity supply and demand at sub-annual time scales.

Table 2
Summary of NDC and LT-LEDS targets for GHG emissions reduction and renewables in the power sector in the Maghreb countries for the years 2030 and 2050.

	GW	Tunisia [8,65]		Algeria [68,69,78]		Morocco [7,70,79,80]	
		2030	2050	2030	2050	2030	2050
Emission reduction	Conditional	−27 % carbon intensity/2010	−68 % GHG emission/BAU	−7 % GHG emission/BAU	–	−18.3 % GHG emission/BAU	−74 % GHG emission/BAU
	Unconditional	−48 % carbon intensity/2010		−22 % GHG emission/BAU		−27.2 % GHG emission/BAU	
RES targets in the power sector		30 % RES-E	80 % RES-E	27 % RES-E	–	52 % RES capacity	80 % RES capacity
RES defined the capacity expansion	Wind onshore	1.6	6.6	5	8.2	4.9	44
	Solar PV	2.6	10.3	13.5	22	2.9	36.7
	CSP	0.2	1.1	2	3.3	2	20
	Hydropower	0.05	0	0		1.4	1.2
	Biomass	0.1	0.1	1	1.6	4	6
Total RES capacities		4.6	18.1	21.5	35	15.2	108

The GHG emissions targets of the Maghreb countries encompass the entire energy system, extending beyond the scope of the power sector. [b] Compared to the Business-as-usual (BaU) scenario, [c] Compared to 2010 emissions, [d] Measure per unit of GDP.

4.2.1. *Data sources*

4.2.1.1. Technology data. The primary technology input data, which are independent of the scenario modelled, are regrouped in Table 3 and Table A-1. The technology data used in the TIMES-MAGe model spans from 2018 to 2050, covering both the calibration year and future projections. The techno-economic parameters of the existing and projected fossil fuel-based and renewable technologies (capital costs, fixed costs, operational life, and efficiency) have been collected from Refs. [58,90–92]. The current installed power plants and the retirement capacities in the Maghreb were sourced from the Global Power Plant Database [93], Harvard Dataverse [94], and IRENA reports [58–61,92,95]. The environmental parameters related to emissions were taken from the EDGAR database [96]. Data according to techno-economic parameters for distribution and transmission technologies was provided by the JRC-EU-TIMES model [97]. The discount rate has been set at 5 % per annum throughout the study period, consistent with the jurisdictions of the Maghreb countries [37,46]. For transparency and reproducibility, all key input data used in the TIMES-MAGe modelling exercise are provided in an open-access Excel file available on Zenodo [98].

Due to the limited availability of region-specific data on interconnector upgrade and reinforcement costs, this study draws on findings from the TEASIMED Masterplan (2020–2022) [75]. The techno-economic parameters for the 6-h battery storage modelled in this study were derived or assumed based on data from the IRENA report [58].

4.2.1.2. Energy resources. The renewable energy potential, production, and capacities of the Maghreb region are assessed through IRENA's GIS-based approach working paper [95,99], respectively. The report employs a Geographic Information System (GIS) approach to estimate and map the geographic potential of renewable energy sources, taking into account factors such as solar radiation, wind speed, topography, land use, and infrastructure suitability.

The estimated maximum technical potential of RES in the Maghreb countries is presented in Table 4. Besides RES, the primary energy supply in the TIMES-MAGe model includes the mining of endogenous fossil resources, as well as imported energy. The prices of fossil energy in the international market are derived from Refs. [89,90] and are regrouped in Table A-2. The model enables electricity trading between the three countries. The associated export prices are not defined exogenously but are generated endogenously by the model, based on the shadow price of electricity in each region and time slice. These prices reflect the marginal cost of electricity supply in the exporting country and are used as the transfer prices for regional electricity exchanges.

The normalized electricity generation of solar PV, onshore wind, and CSP per time slice across the four seasons of the year for the Maghreb countries is illustrated in Figure A (2–4).

4.2.1.3. Electricity demand. Electricity demand stands as a cornerstone in shaping the outcomes of the TIMES-MAGe model. The rate of annual growth in electricity demand in the Maghreb was 5 % during the 2000s. In 2023, the gross electricity demand reached 537 PJ. Mechlia [100] elaborated the methodology for long-term forecasting of electricity demand. Forecasting electricity demand involves various factors, including electricity intensity, GDP growth, and the prices of electricity and fossil fuels.

Population growth and urbanization increase demand, while economic development and industrialization drive higher electricity consumption. Between 2000 and 2021, energy intensity in the Maghreb region rose by 6 %. However, this trend is expected to reverse in the future due to the implementation of energy efficiency measures [45]. According to Morocco's national outlook [41]. The Tunisian perspective, as outlined by the Tunisian Institute for Strategic Studies [101] and for Algeria, based on [45], the annual growth rates of electricity demand from 2020 to 2050 are projected to be 3.5 %, 3.0 %, and 2.8 %, respectively.

The annual electricity demand is disaggregated in normalized load curves as presented in Fig. 5. Because real load curve data were not publicly available for all three countries at the time, the methodology developed by Toktarova et al. was followed [102], from the IRENA report [58], which considers seasonal variations and factors like temperature and air conditioning usage. The approach involves projecting electricity demand at an hourly resolution for all countries using a unified framework. This is achieved by decomposing historical electricity consumption data into a series of sine functions, allowing for the analysis of cyclical patterns.

This electricity demand is linked to the economic growth of Tunisia, Morocco, and Algeria. According to Table 5 GDP in the period 2023–2050, based on their socio-economic scenario of the low-carbon strategy 2050 [7,65], projects an annual increase of 2 % for Tunisia and 3.7 % for Morocco. In the absence of a submitted long-term strategy by Algeria, an assumption was made using the [44] "Ambition Scenarios" indicate an annual growth of 3.5 % for Algerian GDP between 2023 and 2050. The analysis does not account for the demand response to electricity prices.

Additionally, population growth is a crucial factor influencing electricity demand. Population increases by 0.51 %, 0.56 %, and 1.01 % per year between 2023 and 2050 for Tunisia, Morocco, and Algeria, respectively, according to the Medium Scenario of the World Prospects from the United Nations [53].

4.2.1.4. Cross-border electricity trading assumptions. The Maghreb region has maintained interconnected electrical grids since the 1950s, evolving into multiple high-voltage transmission lines among Algeria, Morocco, and Tunisia (Fig. 6). Despite the presence of 400 kV and 220 kV high-voltage transmission lines, cross-border electricity trade

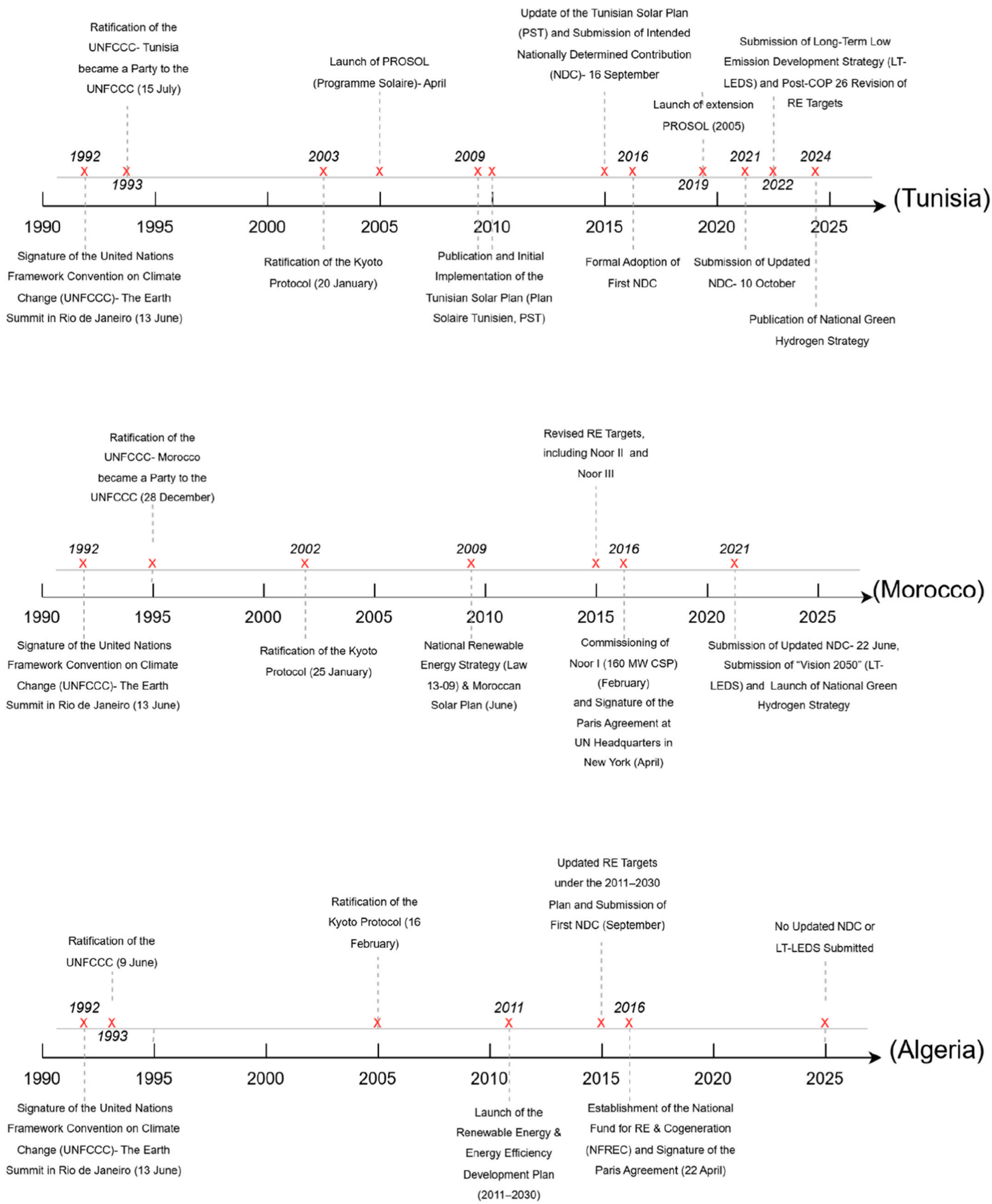

Fig. 3. Evolution of renewable energy strategies and policy milestones in the Maghreb (1990–2025).

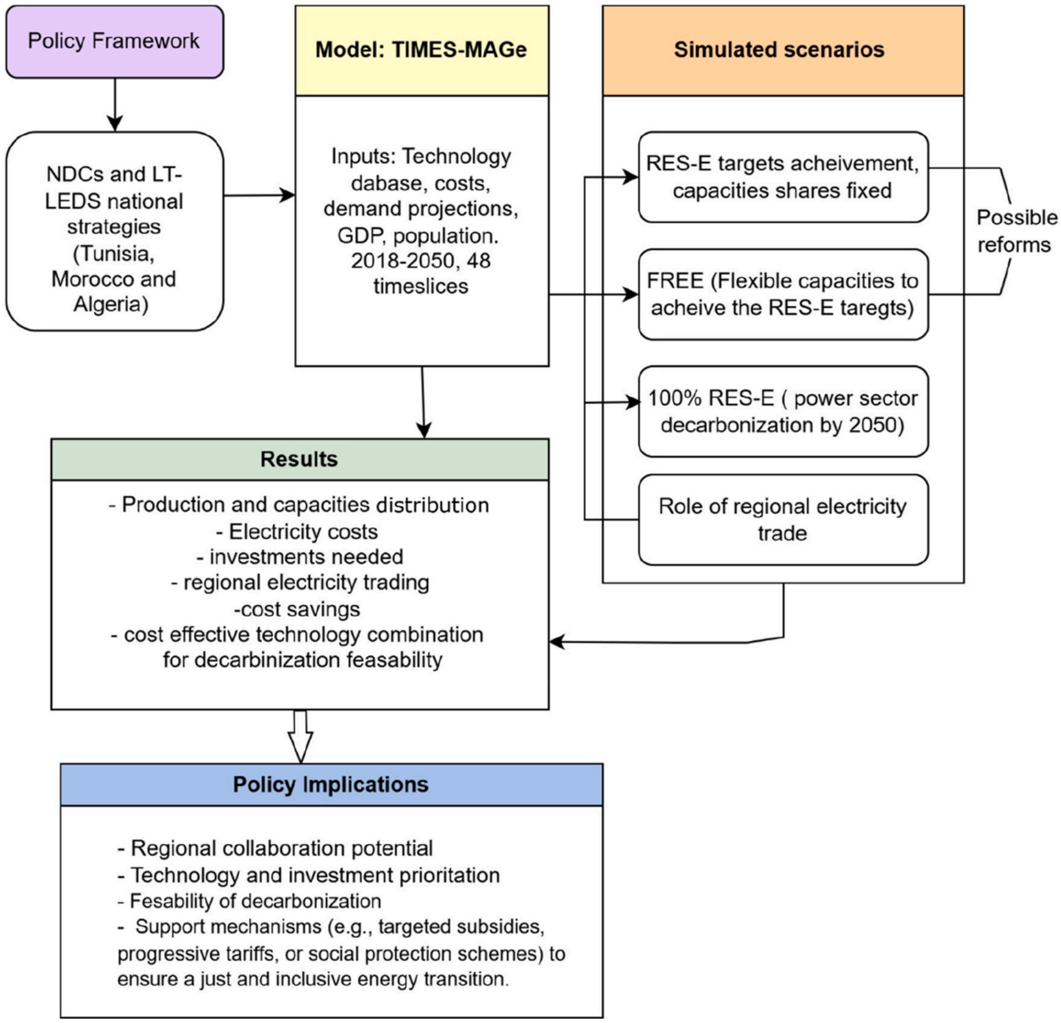

Fig. 4. Scope and analytical framework of the study using the TIMES-MAGE model.

Table 3
Techno-economic parameters of electricity generation technologies in the Maghreb in 2023 and projected costs of renewable energy technologies for selected years up to 2050.

			Capital Cost			Variable Cost	Fixed Cost	Life cycle	Efficiency
	Unit		Million Euro$_{2023}$/GW			Million Euro$_{2023}$/GW/PJ	Million Euro$_{2023}$/GW	Years	%
			2023	2030	2050				
Production	Fuel-fired steam PP	Coal	3050	3050	3050	0.86	91	30	58
		Oil	844	844	844	0.26	24.4	30	42
	Natural Gas		1160	1160	1160	0.48	24.4	30	60
	Onshore Wind		1028	962	855	0.61	42.7	25	100
	Solar PV		789	722	662	0.61	42.7	25	100
	CSP		5800	4200	3355	0.61	49.5	25	100
	Hydropower		2958	2958	2958	2.44	61	80	100
	Biomass		3050	3050	3050	0.42	30.5	25	35
Grid	Upgrade and Reinforcement	Inter-regional	34			–	–	70	95 %
		Cross-border	74.4			–	–		
Batteries Storage			1342	933	629	–	23	8 (3000 cycles)	87 %

remains modest, hindered by political, technical, and market barriers. For instance, Tunisia and Algeria share 1.76 GW of interconnection capacity but currently utilize only 200 MW (around 7 PJ imported in 2023). Morocco and Algeria have a combined capacity of 1.5 GW; however, diplomatic tensions have limited usage to approximately 0.06 PJ in 2020 and effectively halted trade since 2021. Additionally, Morocco is also interconnected with Spain through two 400 kV high-voltage AC interconnectors (totalling 1.4 GW), facilitating around 7 PJ of trade in 2023 and thus connecting the Maghreb to Europe.

In deriving the modelled upper bound for regional electricity trade, we converted each interconnector's capacity into an annual energy flow by applying typical utilization factors and historical peak usage rates. For instance, a 200 MW line operating near its observed maximum could yield around 7 PJ of traded electricity per year. We then scaled these figures proportionally for lines with higher capacities to establish realistic trade limits across all interconnectors.

Table 4
Estimated RES technical potential in the Maghreb [95,99].

	Fossil fuel reserves (PJ)			Renewable energy potential				
				Production (10^3 PJ/year)			Capacities (GW)	
	Coal	Oil	Natural Gas	CSP	Solar PV	Onshore Wind	Solar PV/CSP	Wind
Morocco	452	0.004	52	54.4	54.5	40.6	73	74
Tunisia	0	2.6	242	7.4	16.6	24.6	71	92
Algeria	1906	74	167753	95.5	100	108	54	113

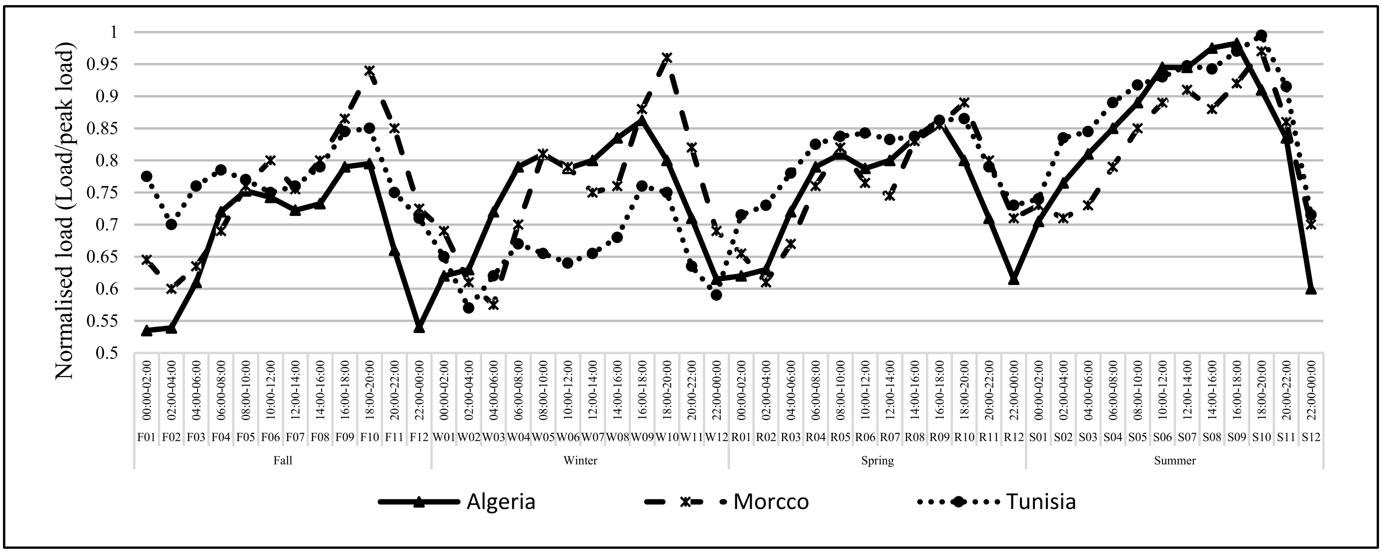

Fig. 5. Normalized load curves on an average day in each season in the Maghreb countries (applied for all years of the modelling period).

Table 5
GDP, population, and electricity demand projections for the Maghreb countries.

		2023	2030	2040	2050
Morocco	GDP (Billion USD 2018)	144.4	186.2	267.8	385.1
	Population (Million inhabitants)	37.1	38.5	40.79	43.6
	Electricity demand (PJ)	119.6	159.7	210.6	280
Tunisia	GDP (Billion USD 2018)	55.7	67.9	66	82.8
	Population (Million inhabitants)	12	12.6	13.3	14
	Electricity demand (PJ)	64.8	91.4	128.9	181.9
Algeria	GDP (Billion USD 2018)	247.6	315	444.3	626.8
	Population (Million inhabitants)	46.1	49.5	54.7	61
	Electricity demand (PJ)	237.2	318.8	428.4	575.8

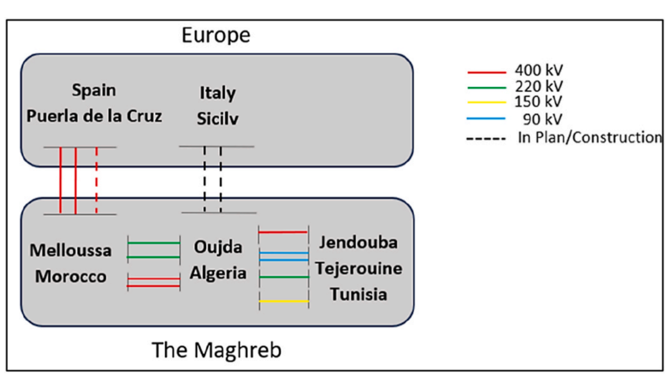

Fig. 6. Scheme of the Maghreb grids and neighboring interconnections adopted from [103].

Table 6
Regional electricity interconnections and modelled upper bounds.

Interconnection	Technical Capacity (GW)			Base year trade (2023)	Modelled Upper bound (PJ)	
	2018	2030	2050		2030	2050
DZR-TUN	1.8	1.55	2.55	12	26	33
MAR-DZR	1.5			0	54	
From Spain to MAR	1.4	2	2	7.2	18.5	18.5

Table 7
Scenarios and assumptions used in the TIMES-MAGe model run.

Scenario	Assumption			
	2030-2050	**TUNISIA**	**MOROCCO**	**ALGERIA**
NDCs/LT-LEDS	**NG**	55% - 17.8%	23% - 10%	25%
	OIL	0.10% - 0	4% - 0	0
	COAL	0	21% - 0	0
	HYDROPOWER	0.08% - 0	12% - 8%	0
	BIOMASS	0.1 % - 0.14 %	0	2%
	ONSHORE WIND	25% - 45.8%	20% - 27%	9%
	SOLAR PV	13.2% - 24.2%	14% - 13%	25%
	CSP	6.6 % - 12.1%	6% -8%	3.6%
FREE		**TUNISIA**	**MOROCCO**	**ALGERIA**
	2030	35%	52%	27%
	2050	80%	80%	N/A

In this study, existing interconnectors among Morocco, Tunisia, and Algeria are maintained and reinforced without expansion. The upper bounds for cross-border electricity trade are set based on the maximum feasible energy flow (taking into account the saturation hours) corresponding to the current and the planned interconnector capacities, as declared in the TEASIMED. Historical import and export data, including those involving Europe, are incorporated to ensure realistic baselines; however, no new intercontinental links are modelled. Instead, the focus remains on national transitions and intra-Maghreb exchanges rather than potential exports beyond the region. Table 6 Summarizes these cross-border trading assumptions.

While the Maghreb region is physically linked to Europe, this study restricts electricity trade modeling to intra-Maghreb exchanges. This reflects current trade baselines and keeps the focus on regional dynamics. Planned interconnections, such as those between Algeria and Italy, are excluded due to their early-stage status. Accordingly, the TIMES-MAGe model is geographically bounded to optimize investment and trade within the Maghreb, without modeling cross-continental flows.

4.3. Scenarios modelled

The three simulated scenarios for the power sector are built upon key assumptions (Table 7), primarily centred on RES-E integration. These scenarios are guided by foundational objectives that define the decarbonization pathways for the power sector in 2030 and 2050, as outlined below.

i. **The NDCs/LT-LEDS scenario determines whether the prescribed renewable capacity** mix can meet the desired targets in terms of RES-E. However, unlike Tunisia and Algeria, Morocco's target is defined in terms of installed capacities rather than electricity generation. To align with this distinct target structure, a linear scaling approach was employed to estimate future electricity production. This method involves proportionally scaling up from the most recent (2023) baseline capacity-to-production ratio, referred to as $baseline_{\_Production}$. By maintaining this ratio, the analysis projects that Morocco will achieve 28 % of RES-E by 2030.

This scenario will provide valuable insights into the feasibility of the proposed strategies, highlight any potential gaps, and assess the reliability of the power sector transition in the Maghreb countries. It also serves as a comparison for the scale of the NDC/LT-LEDS cost-effectiveness when compared to FREE. The NDCs/LT-LEDS targets are regrouped in Table 7.

ii. **FREE: The FREE** scenario defines the desired renewable electricity (RES-E) targets for the Maghreb countries but does not prescribe specific capacity values per technology. Instead, it allows the model the flexibility to select the most cost-effective technologies to achieve the RES-E targets while ensuring that the model does not exceed the predefined national targets. This approach facilitates a comparison between the model-optimized outputs (in terms of production and installed capacities) and the general objectives outlined in national documents.
   - **Tunisia:** While the 2030 installed capacities for each technology are well-documented, specific data on planned capacities per technology (in GW) for 2050 are not publicly available. To address this, we adopted a practical approach: the total planned renewable energy (RE) capacities for 2050, as outlined in the national strategy, served as a reference. To distribute the capacity mix, we assumed that the technology distribution within the total RE capacities in 2050 would reflect the mix set for 2030. We labeled these assumed capacities baselinecap_2030 to clearly identify them

as a reference scenario for Tunisia's future capacity mix. Additionally, in line with Tunisia's national strategy, natural gas capacities were adjusted to ensure they contribute no more than 20 % of total electricity generation by 2050.

- **Morocco***:* For Morocco, while the 2030 target of achieving 52 % of total installed power capacity from renewables is well documented, specific gigawatt figures for 2050 are not publicly disclosed. To address this, we assumed that the annual average growth rate of 9.3 %, observed for renewables between 2020 and 2030, will persist through 2050, maintaining the same capacity mix anticipated in 2030. We named these assumed capacities $baseline_{cap\_2030}$ to identify them as a reference scenario for Morocco's future capacity mix. Under these assumptions, total installed renewable capacities are projected to reach approximately84[3] GW by 2050. In parallel, Morocco anticipates a shift in fossil fuel reliance. As outlined in the [104]Natural gas is expected to gradually replace coal in power generation. Consequently, coal capacities are projected to decline to nearly zero by 2050, while natural gas capacities are expected to increase from 2.4 GW in 2030 to approximately 9.6 GW by 2050.
- **Algeria***:* The scope of our analysis is limited to 2030, as this is the only year for which detailed RES-E targets and strategies are publicly available. The 2030 targets include specific capacities outlined in gigawatts for renewable technologies, reflecting the country's national energy strategy. Unlike Morocco and Tunisia, Algeria has not submitted any targets or strategies for RES-E beyond 2030. Given the absence of data or official commitments for 2050, extending the analysis beyond 2030 would require speculative assumptions that could undermine the reliability of the results.

i. **100-RES:** This scenario envisions a fully renewable-powered electricity system by 2050. The model is designed to optimize the power generation mix by selecting the most cost-effective technologies to achieve this goal. For the year 2030, the model is granted flexibility to choose the cost-optimal solution without constraints, enabling it to reach the country's 2030 RES-E targets.

To avoid confusion, it is essential to clarify that terms such as $baseline_{cap\_2030}$ and $baseline_{\_Production}$ do not represent additional scenarios. Instead, they serve as internal reference assumptions used within the defined scenario framework. These elements support the construction of the three modelled scenarios, NDCs/LT-LEDS and FREE, but are not scenarios themselves.

To evaluate the influence of cross-border electricity trading on the Maghreb's power sector, we conducted two runs of each scenario, NDCs/LT-LEDS, FREE, and 100-RES. In the first set of runs, a static trade assumption was applied, treating each country (Morocco, Algeria, and Tunisia) as an isolated system with regional electricity exchanges fixed at the levels observed in the calibration year (2018). In the second set of runs, we enabled regional electricity trading based on the existing and planned interconnector capacities, allowing for dynamic electricity exchanges between the countries.

This dual approach enabled us to compare the effects of static versus dynamic trade configurations on key metrics, including investment strategies, electricity costs, and the feasibility of deep decarbonization pathways across the Maghreb. It is essential to note that these scenarios were not designed specifically to explore trade policies, but rather to assess how enabling regional collaboration through existing and planned interconnectors affects the overall outcomes of the power sector. The deployment of storage technologies is available as a model option across all scenarios, with inclusion based on their cost-effectiveness.

[3] this includes 40 GW designated for exportation to the European Union [123], which is beyond the scope of this analysis. Therefore, the remaining RE capacities to support the national demand amount to 44 GW.

5. Results and discussion

This section presents the results of the modelled scenarios following a data-driven comparative structure across the Maghreb countries. Detailed numerical results and model outputs supporting the figures and analysis presented in this section are available in the supplementary dataset [105].

5.1. Installed capacities

This section presents a comparative synthesis of installed electricity capacities under the FREE and 100-RES scenarios[4] across the Maghreb region, summarized by country in Fig. 7.

By 2030, the total installed renewable electricity capacity under the FREE scenario reaches approximately 65 GW, closely aligning with national policy targets ($baseline_{cap\_2030}$), with a discrepancy of less than 6 %. Onshore wind emerges as the dominant source, accounting for nearly 73 % of total RES capacity, followed by solar PV at 13 % and CSP at 7 %. This distribution marks a shift away from *the baseline cap* mix anticipated in national plans, where CSP and solar PV held higher shares. In parallel, the 100-RES scenario projects a similar overall capacity in 2030 but reflects a more diversified technology mix, especially in Algeria, where CSP deployment begins to ramp up in preparation for a fully renewable system by 2050. By 2030, when RES-E targets stay below 50 %, the model heavily favours onshore wind over solar PV and CSP. In Tunisia, onshore wind capacity represents 53 % in $baseline_{cap\_2030}$ vis-à-vis approximately 70 % in the FREE scenario of the total RE capacities. In comparison, solar PV represents 27 %–15 % in baselinecap_2030 and the FREE scenario, respectively.

By 2050, both scenarios show convergence in the structure of the renewable electricity mix, although they differ in total installed capacity due to variations in ambition. The FREE scenario results in a total regional RES capacity of approximately 64 GW in Morocco and Tunisia together. Algeria is eliminated from the 2050 target due to the lack of a national target in 2050. In contrast, the 100-RES scenario, which targets the complete decarbonization of the electricity sector in all three countries, requires approximately 130 GW of installed renewable capacity, representing around 30 % of its technical resource potential and following a linear expansion path from the 2030 targets. In both cases, onshore wind retains the largest share, around 55 %, while solar PV accounts for approximately 35–36 %. CSP remains limited to 7–10 % of total capacity. The model's prioritization of onshore wind over solar PV in The Maghreb, aligns same with the insights from Mahlooji et al. [106] which underscores the desirability of wind energy in North Africa for a transition away from carbon-intensive technologies, based on their environmental and economic footprints using the Relative Aggregate Footprint (RAF) indicator.

This capacity mix aligns with the findings of other studies [107–110] Under RES scenarios, favouring diverse RES technologies instead of deploying a single technology, with onshore wind, solar PV, and CSP leading the way for RES installed capacities. However, in Algeria, under the 100-RES scenario, it reaches 22 % due to its dispatchable nature and integration with thermal storage. Hydropower and biomass remain marginal contributors across all scenarios, with technical and environmental constraints limiting their long-term expansion. Hydropower is expected to reach saturation by 2030 and contribute less than 2 % of total installed capacity by 2050, while biomass consistently remains below 1 % due to resource limitations and competing land-use priorities.

National dynamics mirror these regional trends. Morocco and

[4] This comparative assessment focuses only on the FREE and 100-RES scenarios. The NDCs/LT-LEDS scenario is excluded, as it assumes a fixed capacity mix based on national plans rather than optimizing technology choices. Comparing its installed capacities with those derived endogenously would not be methodologically appropriate.

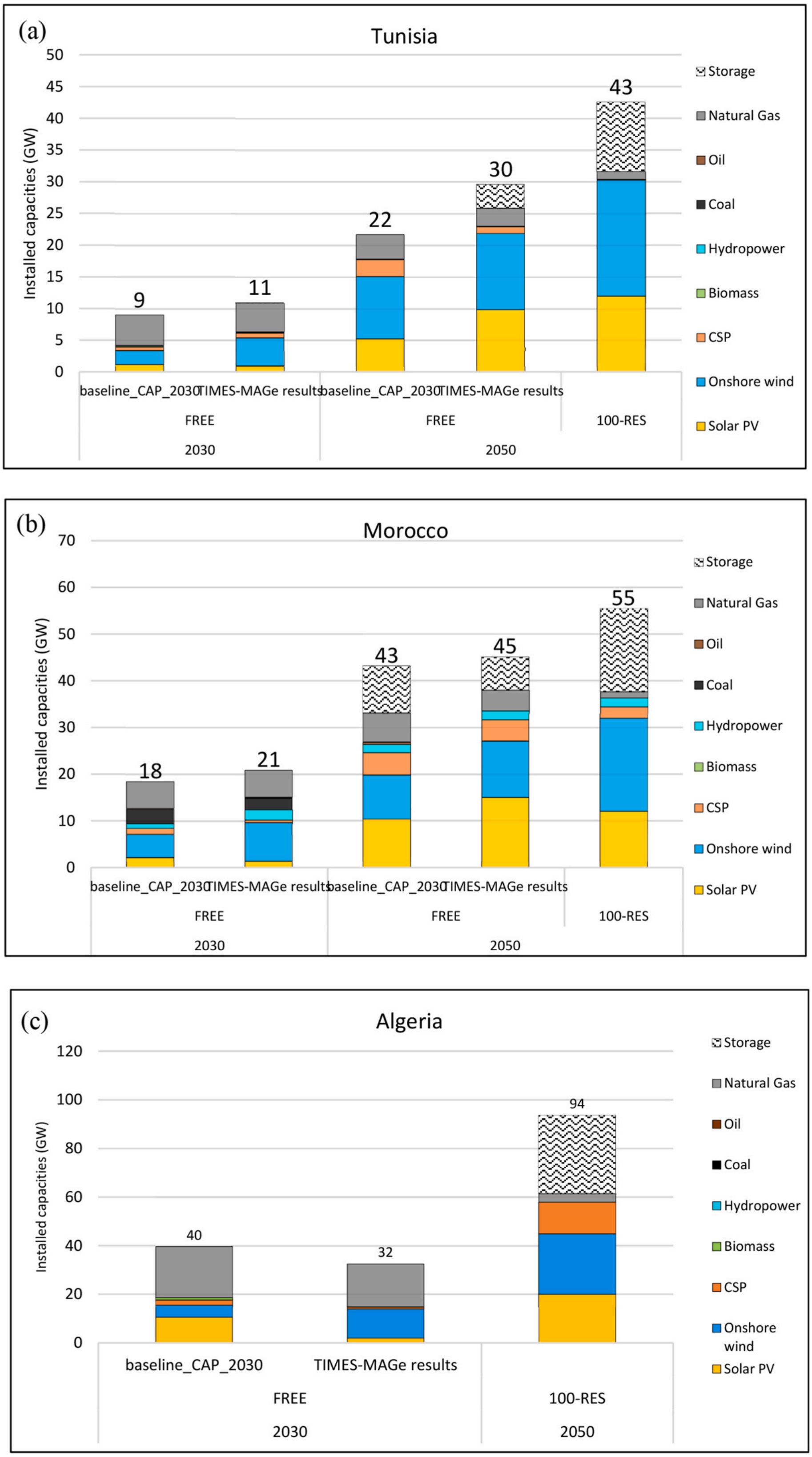

Fig. 7. (a), (b). Installed renewable-electricity capacities by scenario in the Maghreb for 2030 and 2050.
(c). Installed renewable-electricity capacities by scenario in the Maghreb for 2030 and 2050.

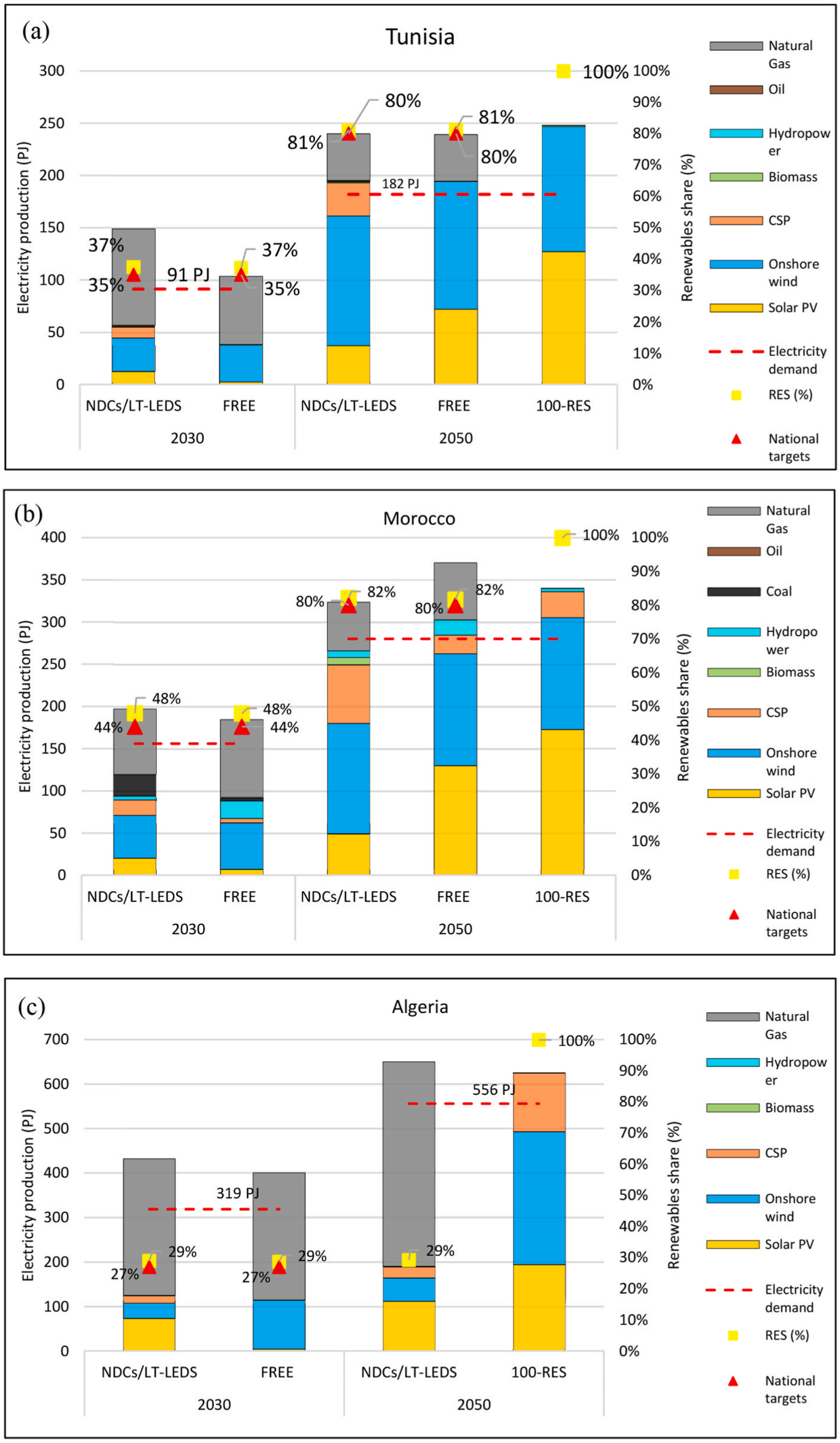

Fig. 8. (a), (b). Electricity generation by scenario in the Maghreb for 2030 and 2050.
(c). Electricity generation by scenario in the Maghreb for 2030 and 2050.

Table 8
Comparison of end-user electricity prices across scenarios for Tunisia, Morocco, and Algeria.

Euro$_{2023}$/kWh		NDCs/LT-LEDS	FREE	100-RES	Elec prices (2013-2023)	Ref
Tunisia	2030	0.14	0.10	0.10	0.036-0.045	[116]
	2050	0.14	0.13	0.28		
Morocco	2030	0.13	0.09	0.09	0.071- 0.087	[117]
	2050	0.13	0.12	0.24		
Algeria	2030	0.12	0.11	0.10	0.023- 0.035	[118]
	2050	0.15	-	0.98		

Tunisia exhibit a strong preference for onshore wind in 2030, comprising over two-thirds of installed RES capacity under the FREE scenario. However, by 2050, solar PV, particularly when combined with battery storage, is expected to experience rapid expansion. In Tunisia, the share of solar PV rises from 15 % in 2030 to 43 % by 2050, while in Morocco it increases from 15 % to 45 % over the same period. CSP is progressively being deprioritized in both countries due to its higher costs, PV's lower costs, which are three to five times cheaper than CSP, and the superior flexibility offered by PV-storage systems. Crucially, once renewables exceed 80 % of electricity production, the model strongly favours PV coupled with batteries over CSP, given that storage can offset the variability of solar power at a lower total cost. Unlike in Algeria, while wind and PV also expand significantly, reaching 12 GW and 15 GW, respectively, by 2050, CSP retains a more prominent role, particularly under the 100-RES scenario. Notably, fossil-based capacity, particularly natural gas, remains significant in Algeria under the FREE pathway, accounting for 44 % of total installed capacity in 2050, underscoring the continued economic and strategic importance of gas in this context.

A critical insight across both scenarios is the emergence of energy storage as an indispensable system component. In the 100-RES scenario, where variable renewables supply the entirety of electricity demand by 2050, approximately 28 GW of storage is required across the region. This capacity is essential for managing intra-day and seasonal variability, ensuring grid stability, and maintaining system reliability without relying on fossil fuel backup. In the FREE scenario, even with lower overall renewables share, storage is still identified as cost-optimal: the model deploys 7 GW in Morocco and 4 GW in Tunisia by 2050. These figures are broadly consistent with Morocco's national strategy, which aims to achieve 10 GW of storage capacity to support its expansion of renewable energy. The alignment between model outcomes and national ambitions reinforces the strategic centrality of storage, not only in fully decarbonized systems but also in transitional, cost-driven pathways.

Taken together, the installed capacity results from both scenarios reveal a consistent technological hierarchy, led by wind and solar PV, with CSP playing a complementary but limited role. While the absolute scale of capacity differs depending on the ambition level, the structural composition of the renewable mix and the reliance on large-scale storage converge. This convergence suggests that policy focus should extend beyond technology selection to ensuring adequate investment in enabling infrastructure to support a stable, reliable, and cost-effective transition to high renewable penetration across the Maghreb.

5.2. *Electricity generation*

Across the three scenarios NDCs/LT-LEDS, FREE, and 100-RES the Maghreb's electricity mix progresses along a common technological hierarchy, yet diverges significantly in the degree of decarbonization achieved by 2050. As illustrated in Fig. 8.

In the policy-constrained **NDCs/LT-LEDS** pathway, national capacity plans are treated as fixed inputs; yet, even under these preset choices, the TIMES-MAGe model projects that all countries will surpass their renewable electricity goals. Tunisia and Algeria each produce about 37 % and 29 % RES-E in 2030, 6.5 percentage points above target, while Morocco attains 44 %, higher than the 28 % yielded by a simple linear scaling of today's fleet. By mid-century, Morocco and Tunisia slightly exceed their 80 % ambitions, while Algeria, still lacking an official 2050 benchmark, reaches one-third of its renewable energy target.

These overachievements stem from the use of up-to-date capacity factors sourced from IRENA rather than the more conservative or unspecified values embedded in national documents [58]. Present-day progress mirrors these modelled advantages: Morocco already has 37 % renewable capacity (as of 2023) and holds a Regulatory Indicator for Sustainable Energy (RISE) score of 74, on par with Sweden and only four points behind China [111]. Tunisia, by contrast, has reached just 4 % RES-E and has seen its Energy Transition Index fall by 7 % since 2012, while Algeria ranks 88th of 115 countries on the same metric and has yet to submit an LT-LEDS document [112].

When technology choice is allowed and only the cost-optimal route to each target is required (the FREE scenario), onshore wind quickly

Table 9
Total Power system investments across scenarios for Tunisia, Morocco, and Algeria.

Billion Euro$_{2018}$		NDCs/LT-LEDS	FREE-REFORMS	100_RES
Tunisia	2030	1.3	1	1
	2050	2.2	1.7	2.4
Morocco	2030	1.2	1.1	1.1
	2050	3.2	2.1	4.5
Algeria	2030	5.3	2.1	2.1
	2050	5.8	-	10.1

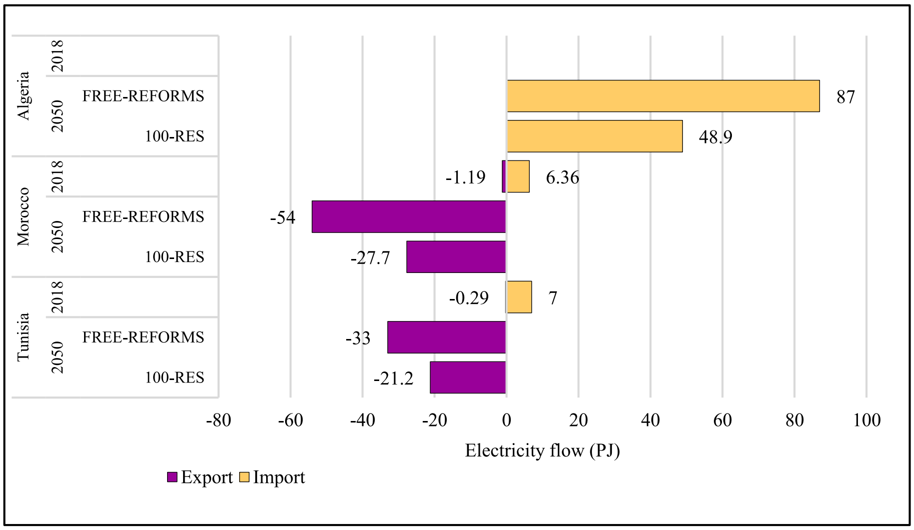

Fig. 9. Gross exports and imports of electricity in Tunisia and Morocco in 2050 vs (2023) values.

becomes the backbone of generation, supplemented by an accelerating contribution from solar PV. By 2050, the two technologies are expected to supply roughly 85 % of Morocco and Tunisia's renewable output, with CSP, hydro, and biomass sharing the remainder. CSP retains a strategic niche in Morocco due to its higher capacity factor; however, its overall share remains below 10 % because PV-battery systems offer similar flexibility at a lower cost. The hydropower potential has already been almost fully exploited [72], so its regional contribution peaks at only 0.7 % of generation by 2050. Climate-driven precipitation declines of 8.5 % could reduce this figure further [3]. Biomass remains below 1 % of capacity throughout the horizon due to limited feedstock, competing land uses, and low conversion efficiency [113]. Crucially, this economically guided mix allows for a sharp contraction of fossil fuel generation without jeopardizing reliability. Gas output in Tunisia falls a quarter below baseline expectations, coal virtually disappears from Morocco's portfolio, and the remaining gas plants operate mainly as flexible reserves.

The 100-RES scenario demonstrates that the region can push further, meeting all electricity demand with renewables by mid-century, while exploiting only about one-third of its wind and solar potential. The hierarchy of resources remains unchanged. Wind delivers the most significant annual energy, PV supplies daytime peaks, and CSP offers dispatchable support, but the supporting infrastructure grows markedly. Around 28 GW of utility-scale storage is required to reconcile hourly mismatches, confirming that flexibility, rather than primary energy, becomes the limiting factor as renewable shares approach 100 %.

5.3. Investment and electricity costs

The evolution of electricity unitary costs for end-users, from historical values [114–116] to projected values by 2050 across all scenarios, as shown in Table 8 provides vital insight into the economic implications of different decarbonization scenarios. In general, higher renewables integration correlates with rising electricity costs, with the 100-RES pathway exhibiting the most significant increase. Between 2030 and 2050, the Maghreb region's electricity unit cost is expected to climb by 23 %, 27 %, and 78 % for Tunisia, Morocco, and Algeria, respectively, under 100-RES, translating into annual growth rates of 1.2 %, 1.4 %, and 4 %, Still, these model results deviate from recent historical trends, in which Tunisia, Morocco, and Algeria observed average annual electricity price growth of 2.5 %, 2.2 %, and 5.1 %, respectively over the period (2013–2023) [117].

Under the 100-RES scenario, electricity unit costs increase steadily between 2030 and 2050 as countries transition from meeting national targets to achieving complete decarbonization of the power sector. Tunisia and Morocco exhibit comparable compound annual growth rates of approximately 4.8 % and 4.7 %, respectively. In contrast, Algeria faces a higher yearly increase of around 11.6 %. These disparities reveal the differentiated economic pressures across the Maghreb and underscore the importance of developing tailored national strategies to manage the cost implications of deep decarbonization. The results align with the conclusions drawn by Ref. [118], indicating a substantial increase in average electricity generation costs.

Compared with the policy-baseline pathway (NDCs/LT-LEDS), Table 9, the cost-optimal FREE scenario delivers substantial savings: average electricity prices fall by 25 % in Morocco and Tunisia and by 10 % in Algeria in 2030, with corresponding reductions of 8 %, 9 % in Tunisia and Morocco, respectively, in 2050. These lower tariffs stem from a parallel decline in capital expenditure, as total power-sector investment across the Maghreb is 29 % lower in 2030 and 35 % lower in 2050 under FREE than under NDCs/LT-LEDS. The picture reverses when the region pursues complete decarbonization. Switching from NDCs/LT-LEDS to the 100-RES pathway raises 2050 electricity prices by 47 % in Tunisia, 45 % in Morocco, and, owing to Algeria's heavy reliance on inexpensive gas, 84 % in Algeria. Achieving 100 % RES-E requires a cumulative spending of roughly €17 billion by 2050, with storage alone absorbing about 14 % of this total. Across all scenarios, aggregate investment for the three countries ranges from €4.2 billion to €17 billion in 2030 and increases by 30–75 % by 2050. FREE remains the least-cost option, while 100-RES is the most capital-intensive, driven by a 49 % expansion in installed capacity and the need for large-scale storage.

These factors increase annualized power system costs by 60 % above the FREE scenario, highlighting the trade-offs between deep decarbonization and economic efficiency in the Maghreb. Given the significant capital requirements for this transition, international funding and strategic partnerships offer viable means to alleviate fiscal burdens, thereby supporting a more sustainable and resilient electricity infrastructure [119].

Similar findings are obtained by Zelt et al. [108], Haller et al. [120], and Dii report [32], and Kost [121] This suggests an unfavorable outlook

Table 10
Impact of regional electricity trading on the Maghreb countries (percentage change compared to static trade.

Scenarios	Country	End-User electricity price	Investments costs	Savings (Annualized total system costs)	Electricity Exports (% of Maximum Interconnector Capacity)
FREE-REFORMS	Tunisia	−14 %	−23 %	33 %	90 %
	Morocco	−13 %	−17 %	28 %	51 %
	Algeria	−8 %	−18 %	−36 %	N/A
100-RES	Tunisia	−58 %	−16 %	4 %	53 %
	Morocco	−47 %	−5.6 %	2 %	56 %
	Algeria	−80 %	7.7 %	2.3 %	N/A

for unitary electricity costs in 100 % renewable energy scenarios, highlighting the high costs associated with a fully renewable energy system in North African countries.

5.4. Regional electricity trade

The introduction of regional electricity trading among the Maghreb countries under the FREE scenario and 100-RES scenarios enables countries to use existing interconnector capacity to trade electricity efficiently. A regionally optimized power system with trading fundamentally transforms the region's energy dynamics, with Tunisia and Morocco emerging as net exporters to Algeria, reversing the historical role of Algeria as the sole exporter in 2023. This paradigm shift is expected to be observed by 2050. It is primarily attributed to Tunisia and Morocco's ambitious RES-E targets (80 % or higher) under the FREE scenario, which drive substantial investments in onshore wind and solar PV, resulting in surplus renewable electricity during periods of peak generation. Compared to static trade scenarios, expanding trade between the Maghreb countries beyond historical trends significantly reduces electricity prices for end-users. Under the FREE scenario by 2050, end-user electricity prices decrease by 0.025 Euro 2023/kWh, corresponding to a reduction of 0.013 Euro 2023/kWh (−14 %) in Tunisia and 0.012 Euro 2023/kWh (−13 %) in Morocco. In the 100-RES scenario, the reductions are even more pronounced, with end-user prices decreasing by 58 % in Tunisia and 46 % in Morocco, despite an increase in investment costs of 23 % and 17 % under FREE-REFORMS and 16 % and 6 % under 100-RES for Tunisia and Morocco, respectively. For Algeria, regional trading represents a cost-optimization strategy, particularly during specific time slices when domestic renewable generation or storage options are less economical (Figure A (5–6)). The gross exports and imports for each country in the results of the FREE scenario and 100-RES are illustrated in Fig. 9.

Enabling the regional trade between the Maghreb countries is characterized by a lower total installed capacity in Algeria by 2050, by 10 % and 2 % under the FREE scenario and 100-RES scenarios, respectively, mostly observed under the CSP capacities, but with an additional 1 GW of batteries, as batteries are highly suited to reduce the day-night variability of solar PV output.

Importing lower-cost electricity from its neighbors results in a notable decrease in electricity costs for end-users by 10 % and 80 % under the FREE scenario and the 100-RES scenario, respectively, compared to the static trade scenario. At the same time, Algeria's total investment costs decrease under trade-enabled scenarios by 18 % (FREE) and 8 % (100-RES). Algeria's heavy reliance on natural gas, its primary endogenous resource, makes a shift to renewables significantly more costly compared to its neighbors with higher RES capacities. The marginal cost of RES-E expansion reveals that Algeria requires €0.47 billion in 2018 to increase its RES-E share by 1 %, compared to just €0.01 billion for Tunisia and €0.05 billion for Morocco in the same year. This highlights Algeria's higher investment intensity for renewable energy transition, reflecting its lower baseline capacity and dependence on fossil fuels.

Regional trade capacities represent 2.5 % and 2 % of the total installed technology capacities in Morocco and Tunisia, respectively, under the FREE and 100-RES scenarios. Additionally, the electricity exported by Tunisia and Morocco accounts for 71 % and 54 % of the maximum allowable flow of the existing interconnectors by 2050 under the FREE and 100-RES scenarios, respectively, demonstrating the effective utilization of regional infrastructure.

To better illustrate the impact of regional electricity trading among the Maghreb countries, Table 10 Summarizes the key changes observed under the FREE and 100-RES scenarios by 2050 compared to the static trade scenario.

This integration of resources reduces the annualized total system costs across the net exporter countries, resulting in savings for Morocco and Tunisia. Under the FREE scenario, Morocco achieves savings of € 1.85 billion in 2018, while Tunisia saves € 0.24 billion in 2018. These savings stem from revenue from electricity exports, which allows both countries to optimize their renewable investments. Similarly, under the 100-RES scenario, Algeria benefits from a reduction in marginal electricity costs by importing lower-cost renewable electricity during periods of high demand, resulting in system-wide savings of €0.3 billion in 2018, creating a win-win scenario.

6. Conclusion

In recent years, Maghreb countries committed to the Paris Agreement goals, outlining NDCs and LT-LEDS. However, the feasibility and cost-effectiveness of proposed measures, especially for ambitious power sector targets, remain uncertain.

This paper provides a valuable resource for researchers, planners, and policymakers by evaluating the viability and economic implications of achieving a 100 % renewable electricity sector in Morocco, Tunisia, and Algeria by 2050. Using TIMES-MAGE, a country-specific linear optimization model, we assess national and regional decarbonization pathways under both constrained and flexible technology scenarios. While assuming a limited and historically grounded electricity trade, the study also explores the benefits of optimizing regional electricity exchanges via existing and planned interconnectors.

This study analyses Maghreb's renewable energy targets (updated NDCs and LT-LEDS) and examines the viability of achieving a 100 % renewable power sector by 2050 at both national and regional levels. It develops TIMES-MAGe, a linear optimization interconnected country-specific power sector model, to evaluate the cost-effectiveness of decarbonization options, considering technology aspects and resource potentials. Scenarios assume a limited and historic electricity trade among countries to focus on the countries' endogenous potential to achieve their targets and the possibility of optimizing the power sector across the entire region through cost-effective electricity exchanges limited by existing and planned interconnections.

The study addresses five core questions, examining.

- To what extent do the planned capacities in NDCs fulfil renewable electricity targets?
- Is there potential for optimizing predefined capacity plans to enhance the cost-effectiveness of achieving national RES goals in the Maghreb countries?
- What is the potential of the Maghreb region to achieve 100 % RES-E, considering the technological capabilities and the potential resources?
- What are the total costs, including investments, and the resulting electricity unitary cost under different power sector scenarios?
- What are the trade-offs of enabling regional electricity trade in the Maghreb?

Results show that all three countries can exceed their RES-E targets under modelled scenarios. Onshore wind and solar PV emerge as the least-cost technologies, aligned with national potentials. In the cost-optimal free scenario, electricity prices are projected to fall by 25 % in Morocco and Tunisia, and by 10 % in Algeria, by 2030 relative to policy-constrained pathways.

However, complete decarbonization under the 100-RES scenario requires significant investment in approximately 130 GW of installed renewable capacity and 28 GW of storage by 2050, nearly double the capacity of the FREE scenario. Wind and PV account for 90 % of the capacity mix, while CSP plays a complementary role. The associated cumulative investment reaches €17 billion, of which 14 % is dedicated to storage. Electricity prices are projected to increase by 47 % in Tunisia, 45 % in Morocco, and 84 % in Algeria by 2050 under this scenario, highlighting the economic trade-offs associated with deep decarbonization.

Regional electricity trading emerges as a key enabler of cost optimization. It reduces Algeria's investment needs by 18 % (FREE) and 8 % (100-RES), while allowing Morocco and Tunisia to export up to 71 % and 54 % of their interconnector capacities, respectively. These dynamics reverse historical patterns, positioning Morocco and Tunisia as net exporters and providing Algeria with cost-effective imports during periods of peak demand. These findings position the study as a catalyst for renewed policy dialogue and scholarly attention on Maghreb-wide renewable energy integration, an essential yet underexplored pillar of the region's decarbonization strategy.

This study identifies cost-effective technology pathways, such as expanded solar PV, Wind, and CSP deployment, battery storage integration, and regional electricity trade, that can inform updates to Morocco, Tunisia, and Algeria's NDCs, aligning national planning with long-term decarbonization goals.

While the study underscores the strategic importance of renewable expansion, storage deployment, and regional cooperation, it also reveals significant limitations.

- Results rely on scenario modeling (TIMES-MAGe) without empirical validation.
- Socioeconomic and employment impacts are not captured.
- Data inconsistencies across sources may affect cross-country comparability.
- The study is limited to the power sector, excluding sector coupling, Power-to-X, CCS, and CCU.
- Institutional and infrastructural trade barriers are not fully addressed.

Despite these limitations, the study's strength lies in establishing a robust modelling framework tailored to Maghreb countries. It enables a detailed exploration of national RES targets, investment needs, and regional coordination opportunities to inform future energy planning and policy reform.

Looking ahead, future research should adopt more integrated approaches, such as the Smart Energy Systems (SES) framework, that capture cross-sectoral dynamics between electricity, heating, cooling, and transport. Further studies should assess the potential role of CCS and CCU, particularly in fossil-fuel-dependent contexts, and evaluate the integration of green hydrogen with electricity systems to unlock broader decarbonization pathways for the region.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.esr.2025.101918.

Maghreb's renewable power future for climate mitigation: insights from the TIMES-MAGe Model-APPENDIX A

This appendix presents the context of The Maghreb's power sector, the data fed into the model described in section.3, and the results of the Business-as-usual scenario from section.4.

1. Reference energy system

Until 2018, there are 130 power plants in The Maghreb countries, categorized into 9 distinct types, coal, diesel and natural gas, i.e., fossil power plants and renewable energy technology such as, hydropower, biomass, photovoltaic, concentrated solar power (CSP), and wind onshore (Figure A-1).

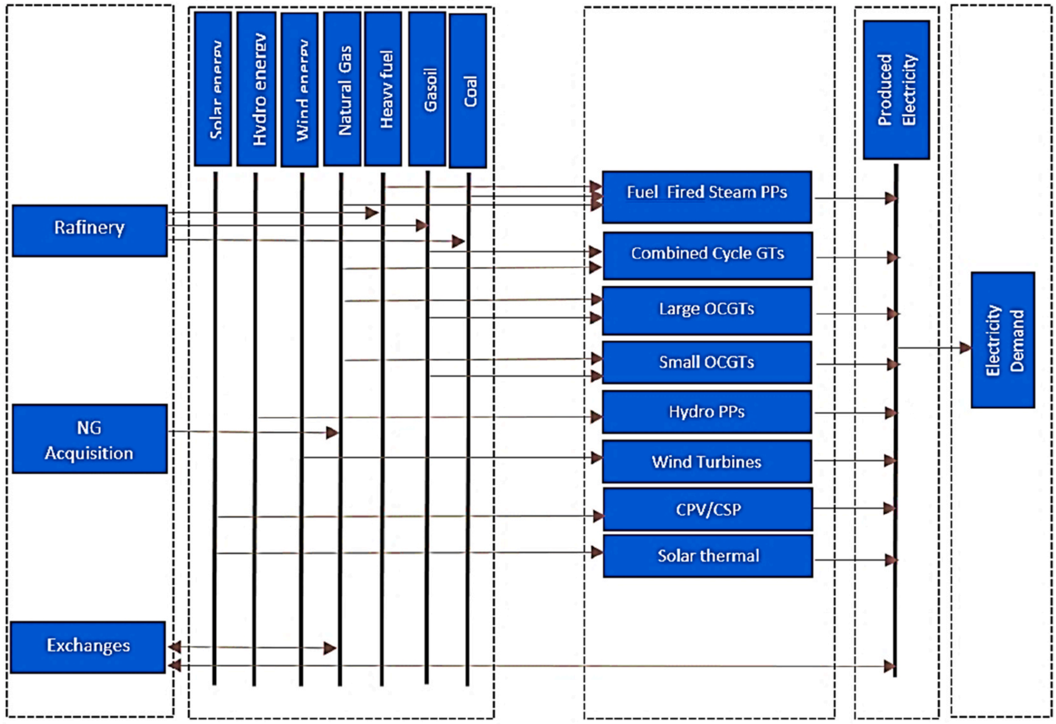

Fig. ure A-1. Simplified scheme of The Maghreb electrical power system in TIMES-MAGe.

2. Technology data

Table A-1
List of the Existing and retirement installed capacities in The Maghreb countries

		Existing installed capacities 2018			Retirement capacities 2030 [a]			Retirement capacities 2050		
GW		Algeria	Morocco	Tunisia	Algeria	Morocco	Tunisia	Algeria	Morocco	Tunisia
Solar		0.3	0.7	0.05	0.3	0.51	0.134	0.058	0	0.050
Wind		0	1.2	0.24	0	1.07	0.15	0	0	0
Natural Gas		19	2.6	4.8	6.49	2.6	3.2	2.5	0	0
Hydropower		0.12	1.7	0.06	0.12	1.7	0.06	0	1.3	0.035
Fuel fired steam	Coal	–	4.3	–	–	3.8	–	–	0	–
	Oil	0.35	0.3	0.02	0.35	0.116	–	0	0	–

[a] The retirement capacities are calculated based on the starting year and the lifetime of each power plant.

Table A- 2
Fossil Fuel Prices, Endogenous Resource Mining Costs, and Fossil Fuel Imports Costs for The Maghreb Region

	Commodity	Fuel Price (USD_{2018}/GJ)				
		2020	2025	2030	2040	2050
Imports	Crude Oil	12.2	12.76	14.27	16.9	19.53
	Biomass	1.76	1.76	1.76	1.76	1.76
	Coal	5.1	5.3	5.5	5.9	5.9
	Heavy fuel oil	8.87	9.28	10.38	12.29	14.2
	Light Fuel Oil	14.7	15.4	17.2	20.4	23.6
	Natural Gas	8.6	9.5	10.3	11.0	11.0
Mining, endogenous resources	Crude Oil	11.09	11.60	12.98	15.36	17.75
	Biomass	1.6	1.6	1.6	1.6	1.6
	Coal	3.4	3.5	3.6	3.8	3.8
	Heavy fuel Oil	–	–	–	–	–
	Natural Gas	7.1	7.8	8.5	9.9	9.9

3. Capacity factor

The capacity factors for each time slice of Solar PV, CSP, and onshore wind are presented in Figure A-2, Figure A-3, and Figure A- 4 below, respectively. Country-specific hourly capacity factors for Solar PV, wind, CSP and hydropower technologies in The Maghreb countries were sources from Renewable ninja [111], PVGIS [112,122] , IRENA [42,83] and the Plexos-world 2015 model dataset [82] and rearranged to the time resolution required by the TIMES-MAGe model (48 time slices as previously described) for all the years in the projections and under all scenarios. We calculated the capacity factors for each time slice by dividing the total electrical output by the maximum possible output for that period. The Renewable ninja, PVGIS and IRENA provide data for 2018 unlike Plexos-world 2015 which provide data until 2015 for CSP and 15-year average monthly capacity factors for hydropower.

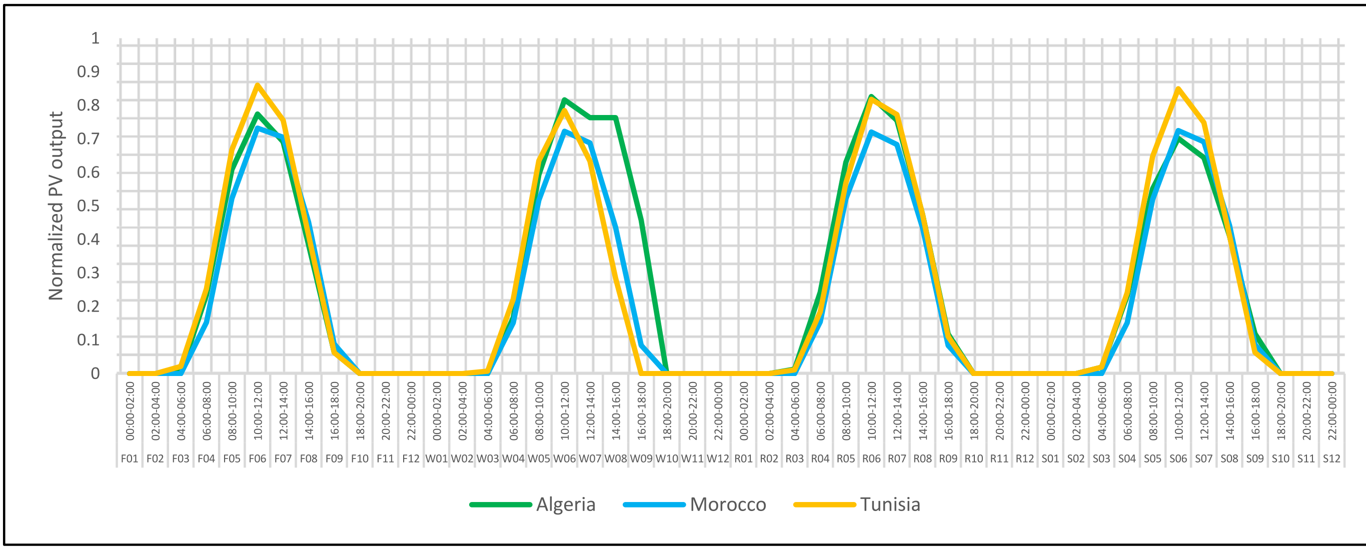

Fig. ure A-2. Normalized Solar Photovoltaic Generation Across Four Seasons and 48 Time Slices for Tunisia, Morocco, and Algeria.

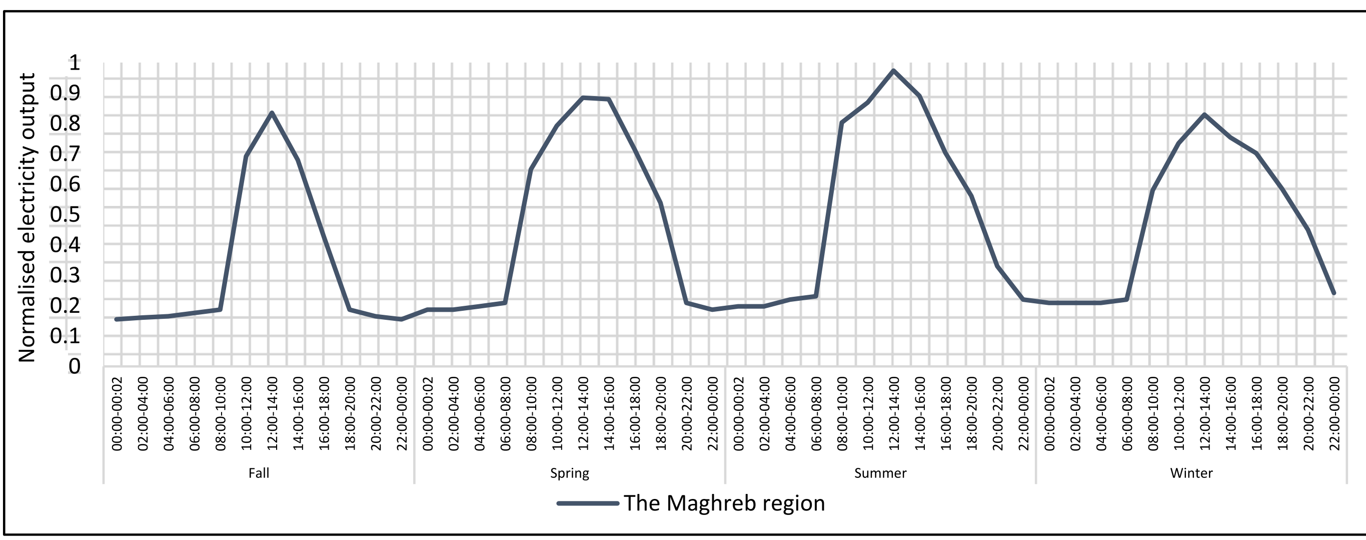

Fig. ure A- 3. Normalized Concentrated Solar Power (CSP) Generation Across Four Seasons and 48 Time Slices in the Maghreb Region.

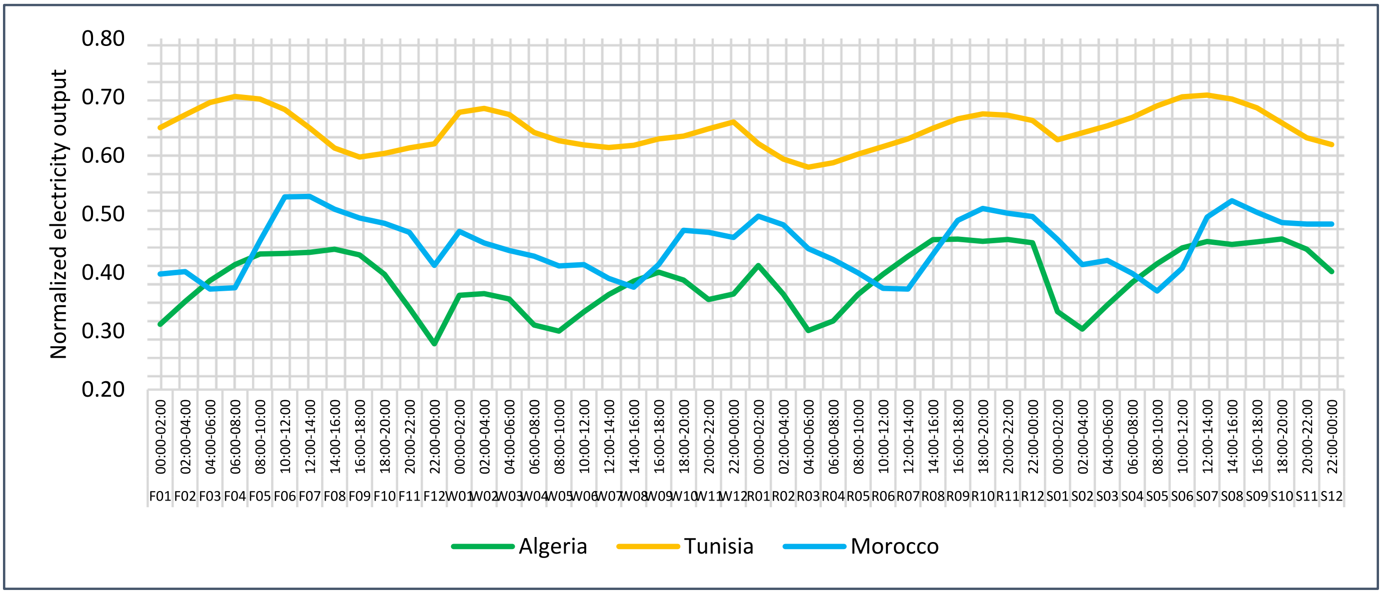

Fig. ure A- 4. Normalized wind energy Generation Across Four Seasons and 48 Time Slices for Tunisia, Morocco, and Algeria.

4. Regional trading

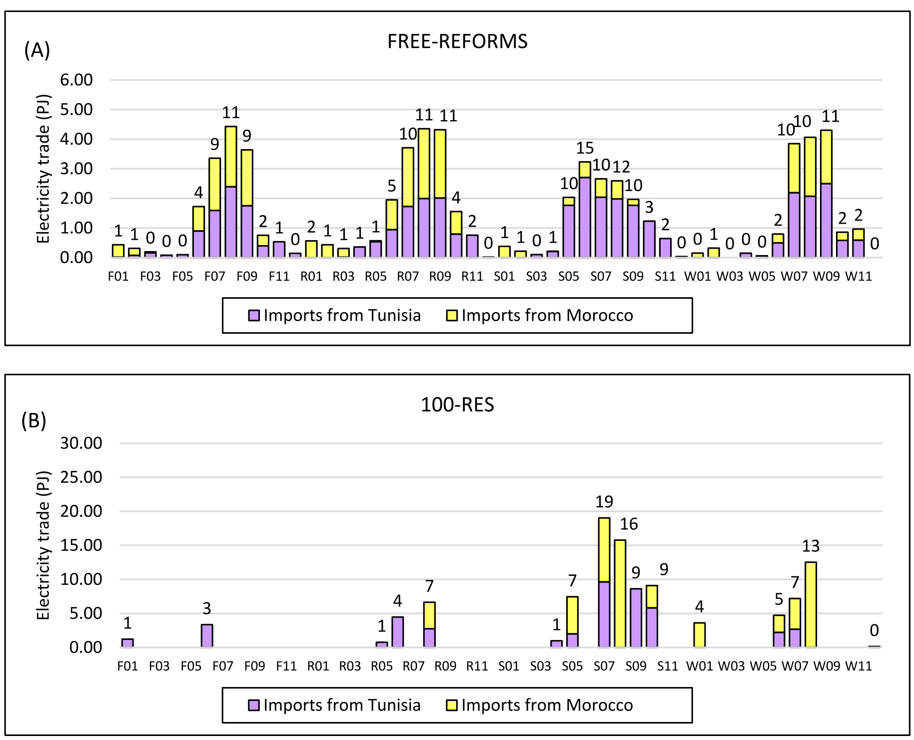

Fig. ure A-5. Distribution of Regional Electricity Trading by Time Slice: (A) FREE Scenario and (B) 100-RES Scenario.

The analysis of Algeria's electricity imports from Morocco and Tunisia reveals a clear temporal pattern that aligns closely with the country's demand profile and seasonal electricity consumption trends. Most of Algeria's reliance on imports occurs during the 14:00–18:00 window in Autumn (F), Spring (R), and Winter (W), which corresponds to mid-to late-afternoon periods of elevated demand driven by commercial, residential, and industrial activities. In Summer (S), Algeria's imports reach their maximum levels, with peaks of up to 15 PJ observed during specific time slices between 12:00 and 18:00. This intensified reliance on imports in summer is likely tied to higher electricity demand and solar PV production in exporting countries. Annually, electricity imports accounts for between 5 % and 10 % of total consumption, achieving a major dependency in summer afternoon (20 %, and 58 %) UNDER FREE scenario and 100-RES scenarios respectively, creating geopolitical challenges due to the country reliance on external energy sources.

Data availability

Data will be made available on request.

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